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FINDING NEURONS IN A HAYSTACK: CASE STUDIES WITH SPARSE PROBING

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ABSTRACT

Despite rapid adoption and deployment of large language models (LLMs), the internal computations of these models remain opaque and poorly understood. In this work, we seek to understand how high-level human-interpretable features are represented within the internal neuron activations of LLMs. We train k-sparse linear classifiers (probes) on these internal activations to predict the presence of features in the input; by varying the value of k we study the sparsity of learned representations and how this varies with model scale. With $k = 1$, we localize individual neurons that are highly relevant for a particular feature and perform a number of case studies to illustrate general properties of LLMs. In particular, we show that early layers make use of sparse combinations of neurons to represent many features in superposition, that middle layers have seemingly dedicated neurons to represent higher-level contextual features, and that increasing scale causes representational sparsity to increase on average, but there are multiple types of scaling dynamics. In all, we probe for over 100 unique features comprising 10 different categories in 7 different models spanning 70 million to 6.9 billion parameters.

1 Introduction

Neural networks are often conceptualized as being flexible "feature extractors" that learn to iteratively develop and refine suitable representations from raw inputs [1, 2]. This begs the question: what features are being represented, and how? Probing is a standard technique used to study if and where a neural network represents a specific feature by training a simple classifier (a probe) on the internal activations of a model to predict a property of the input [3] (e.g., classifying the tense of a verb based on the activations of a specific layer). Because models are parameterized as a series of dense matrix multiplications and elementwise nonlinearities, a natural intuition is that features are represented as linear directions in activation space [4] and are iteratively combined to synthesize increasingly abstract features using linear combinations of previously computed features [5]. To zoom in on these dynamics in modern transformer language models, we propose *sparse probing*, where we constrain the probing classifier to use at most k neurons in its prediction; we probe for over 100 features to precisely localize relevant neurons and elucidate broader principles of how to study and interpret the rich structure within LLMs.

In the simplest case, one might hope there exists a 1:1 correspondence between features of the input and neurons in a network—that for the correct feature definition, a 1-sparse probe would be sufficient. Although the literature contains many examples of such seemingly *monosemantic neurons* [6, 7, 5, 8, 9, 10], an obvious problem arises when a network has to represent more features than it has neurons. To accomplish this, a model must employ some form of compression to embed n features in $d < n$ dimensions. While this *superposition* of features enables more representational power, it also causes loss-increasing "interference" between non-orthogonal features [4]. Recent works in toy models demonstrate that this tension manifests in a spectrum of representations: dedicated dimensions or neurons for the most prevalent and important features with increasing levels of superposition—and hence decreasing levels of sparsity—for the long tail of rarer or less important features [4, 11].

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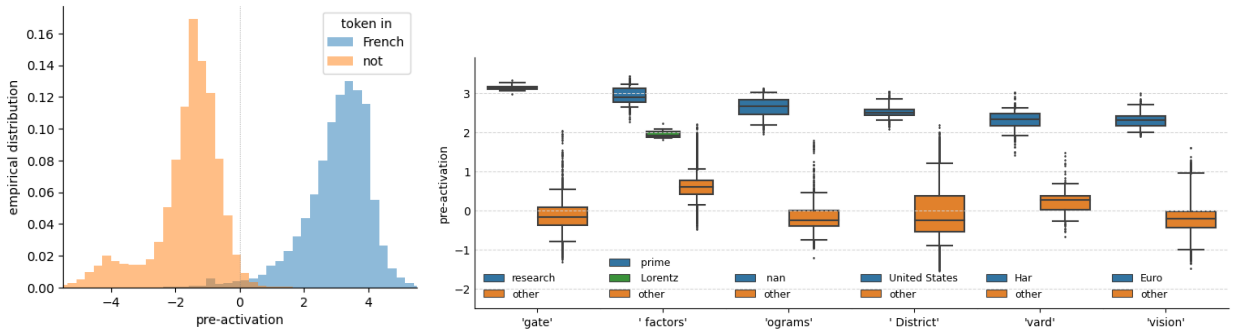

(a) A monosemantic French neuron

(b) A polysemantic neuron activating on six unrelated n-grams

Figure 1: Neurons that respond to single features (left) can be understood independently, in contrast to polysemantic neurons (right) that activate for many unrelated features (in this case, the occurrence of specific multi-token spans).

As with any conceptual insight, these results raise more questions than they answer: To what extent do these observations transfer to full scale language models? What kinds of features do or do not appear in superposition? At what scale? How do we reliably find and verify such (feature, neuron) pairs? We leverage sparse probing to systematically study such questions, finding clean examples of both monosemantic neurons and superposition "in the wild." Better understanding—and potentially resolving—superposition is on the critical path to ambitious full-model interpretability because a logical consequence of superposition is the presence of *polysemantic* neurons that activate for a large collection of seemingly unrelated stimuli (Figure 1) [10, 12]. Such tangled representations undermine our ability to decompose networks into independently meaningful and composable components, thwarting existing approaches to reverse-engineer networks [5].

In addition to being especially well suited to studying superposition, our sparse probing methodology addresses several shortcomings identified in the probing literature [3, 13, 14, 15]. By leveraging recent advances in optimal sparse prediction [16, 17], we are able to prove optimality of the k-sparse feature selection subproblem (for small k), addressing the conflation of ranking quality and classification quality raised in [13]. Second, by employing sparsity as an inductive bias, our probes maintain a strong simplicity prior and are more capable of precise localization of important neurons. This precision enables a more granular level of analysis illustrated throughout our case studies. Finally, the lack of capacity inhibits the probe from memorizing correlation patterns associated with the feature of interest [14], offering a more reliable signal of whether this feature is explicitly represented and used downstream [15].

In the first part of the paper, we outline several variants of sparse probing, discuss the various subtleties of applying sparse probing, and run a large number of probing experiments. In particular, we probe for over 100 unique features comprising 10 different categories in 7 different models spanning 2 orders of magnitude in parameter count (up to 6.9 billion). The majority of the paper then focuses on zooming in [5] on specific examples of general phenomena in a series of more detailed case studies to demonstrate:

- There is a tremendous amount of interpretable structure within the neurons of LLMs, and sparse probing is an effective methodology to locate such neurons (even in superposition), but requires careful use and follow-up analysis to draw rigorous conclusions.
- Many early layer neurons are in superposition, where features are represented as sparse linear combinations of polysemantic neurons, each of which activates for a large collection of unrelated n-grams and local patterns. Moreover, weight statistics combined with insights from toy models suggest that the first 25% of fully connected layers employ substantially more superposition than the rest.
- Higher-level contextual and linguistic features (e.g., `is_python_code`) are seemingly encoded by monosemantic neurons, predominantly in middle layers, though conclusive statements about monosemanticity remain methodologically out of reach.
- As models increase in size, representation sparsity increases on average, but different features obey different dynamics: some features with dedicated neurons emerge with scale, others split into finer grained features with scale, and many remain unchanged or appear somewhat randomly.

2 Related Work

Probing Originally introduced by [18], probing is a standard technique used to determine whether models represent specific features or concepts [3]. For language models in particular, there exist many probing studies showing the rich linguistic representations learned by models [19, 20, 21], contributing to the broader field of "BERTology" [22]. Particularly relevant to our work are investigations on the role of individual neurons [7, 23, 24, 25] and the identification of sparsely represented features [26, 27]. While probing in general has a number of limitations [14, 28, 15, 29, 30], our work seeks to address, at least partially, the shortcomings most associated with probing individual neurons [13]. Methodologically, sparse probing is situated within the broader category of probing methods [31, 32, 33, 34], which itself is just one paradigm among other localization techniques [35, 36, 37, 38, 39, 40].

Mechanistic Interpretability Philosophically, our work is motivated by the growing field of mechanistic interpretability (MI) [5, 41, 42]. MI concerns itself with rigorously understanding the learned algorithms (circuits) utilized by neural networks [43, 44, 45, 46] in the hope of maintaining oversight and diagnosing failures of increasingly capable models [47]. Although research on reverse-engineering specific neurons in LLMs is limited, [48] proposed—and later refined [49]—the hypothesis that feed-forward layers function as key-value memories, responding to specific input features (keys) and updating the output vocabulary distribution accordingly [50].

Superposition Perhaps the most significant obstacle to interpreting neurons in LLMs, and consequently, the success of MI as a whole, is the phenomenon of superposition. As first observed by [51], superposition involves compressing multiple features into a smaller number of dimensions [4]. Recent research has investigated when superposition occurs [4, 11], how to design models to have less of it [52, 10], and how to extract features in spite of it [53]. Many of the underlying mathematical intuitions rely on prior work on compressed sensing [54]. Moreover, similar questions have been studied elsewhere in machine learning within the disentanglement literature [1, 55, 56].

Connections to Neuroscience In a promising demonstration of consilience, our previous discussion of superposition has striking analogues with coding theory from biological neuroscience—the study of how neurons in the brain map to sensory stimulus [57, 58, 59]. On one extreme, local coding theory posits the existence of "monosemantic" biological neurons which respond to a very specific stimulus (e.g., pictures of Jennifer Aniston [60]). Superposition is then analogous to sparse coding where a subset of neurons encode some feature about the input [58, 61]. Finally population coding theorizes the responses of a whole brain region are relevant, analogous to the computations of a full layer being required to compute or represent a feature [62, 63].

3 Sparse Probing

3.1 Preliminaries

Transformers We restrict our scope to transformer-based generative pre-trained (GPT) language models [64] that currently power the most capable AI systems [65]. Given an input sequence of tokens $x = [x_1, \ldots, x_t] \in \mathcal{X}$ from the vocabulary, a model $\mathcal{M} : \mathcal{X} \to \mathcal{Y}$ outputs a probability distribution over the token vocabulary $\mathcal{V}$ to predict the next token in the sequence. $\mathcal{M}$ is parameterized by interleaving L multi-head attention (MHA) layers and multi-layer perceptron (MLP) layers. The central object of the transformer is the residual stream, the sum of the output of all previous layers. Each MHA and MLP layer reads its input from the residual stream and writes its output by adding it to the stream. MHA layers are responsible for moving information between token positions, MLP layers apply pointwise nonlinearities to each token independently and therefore perform the majority of the feature extraction, and the residual stream acts as the communication channel between layers. In this study, we are primarily concerned with the MLP layers (accounting for $\approx \frac{2}{3}$ of total parameters). Ignoring the MHA layer, the value of the residual stream (also referred to as the hidden state) for token t at layer ℓ after applying the MLP is

$$h_t^{(\ell)} = h_t^{(\ell-1)} + W_{\text{proj}}^{(l)} \sigma \left(W_{fc}^{(l)} \gamma \left(h_t^{(l-1)} \right) + b_{fc}^{(l)} \right) + b_{\text{proj}}^{(l)}$$

where W_{proj} , W_{fc}, $b_{\text{proj}}^{(l)}$, $b_{\text{fc}}^{(l)}$ are learned weight matrices and biases, γ is a normalizing nonlinearity, and σ is a pointwise rectifying nonlinearity (in our case GeLU). For a more detailed mathematical explanation of the transformer architecture we refer the reader to [41].

Probing is a technique to localize where specific information resides in a model by training a simple classifier (a probe) to predict a labeled feature of the input using the internal activations of the model [3]. More formally, we require a tokenized text dataset $X \in \mathcal{V}^{n \times T}$ (where n is the number of sequences and T is the context length) and an associated